Aberration Analysis of Reflecting and Refracting Optical Systems

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I. Introduction

The purpose of this laboratory is to use optical design software, ATMOS [2] and Zemax OpticStudio, to measure how the image quality of a system depends on its geometry. Three families are traced: single mirrors, spherical and parabolic, over a range of focal ratios; the classical two-mirror telescopes; and singlet lenses.

Each configuration is judged with the same three tools: the aberration curves, which say what is wrong with the wavefront; the spot diagram, which says how large the image of a point source is; and the modulation transfer function, which says how much contrast survives at each spatial frequency.

II. Theoretical Framework

A. Aberrations

The wavefront aberration W is the departure of the real wavefront from a sphere centred on the image point, measured in waves. Rays travel normal to the wavefront, so the transverse ray aberration ε, the distance between where a ray lands and where the paraxial ray lands, is proportional to the slope of W [1]. Ray fans plot ε and OPD fans plot W; both describe the same defect.

To third order each aberration carries its own power of the normalized pupil radius r and field height β: spherical aberration goes as r^4 and does not depend on the field, coma as βr^3, and astigmatism as $\beta^2 r^2$. For a mirror of diameter D and radius $R = 2f$ at infinite conjugate, the transverse spherical aberration of a sphere and the peak-to-valley wavefront error at best focus are

$$\mathrm{TSA} = \frac{D}{64N^2}, \qquad W_{PV} = \frac{D}{2048\,N^3}, \tag{1}$$

with $N = f/D$ the focal ratio; both vanish for a paraboloid, $K = -1$. Coma does not depend on the conic constant when the stop is on the mirror, and is measured by the offence against the sine condition

$$\mathrm{OSC} = \frac{1}{16N^2}, \tag{2}$$

which becomes a linear blur through $\mathrm{Coma}_{tan} = 3\,\mathrm{OSC}\,h'$ with $h' = f\tan\theta$ the image height.

B. Spot diagram

The spot diagram is the set of intercepts in the image plane of rays uniformly sampling the pupil. It is purely geometrical, so it is meaningful only against the size diffraction imposes even on a perfect system, the Airy disc,

$$d_{\mathrm{Airy}} = 2.44\,\lambda N. \tag{3}$$

A system whose spot fits inside the Airy disc is diffraction limited. Since d_{Airy} grows with N, configurations are compared through the ratio between the RMS spot radius and the Airy radius, not the spot size alone.

C. Modulation transfer function

The MTF is the contrast transferred as a function of spatial frequency, and is the modulus of the Fourier transform of the point spread function. An aberration-free system has an MTF limited only by diffraction, falling monotonically to zero at

$$\nu_c = \frac{1}{\lambda N}. \tag{4}$$

Aberrations depress the curve below that ideal. A strongly aberrated system can show the curve reaching zero and rising again, which is spurious resolution: contrast returns with the fringes reversed.

III. Equipment and Method

TABLE I: Tools used in the analysis.

Item	Use
Zemax OpticStudio	ray tracing, spot diagrams, MTF
ATMOS v8.96	cross-check of aberration coefficients
Personal computer	simulation and data reduction
Schott glass catalogue	refractive indices of the lenses

All systems use the same aperture, $D = 200\,\mathrm{mm}$, so that the spot diagrams of different configurations are directly comparable. The reference wavelength is 550 nm for the mirror systems, which are achromatic by construction; the lenses are evaluated at 450, 550 and 650 nm. The stop is on the first optical surface of every system. The single mirrors are sampled at 0, 0.5, 1 and 2°, the telescopes at 0, 0.25 and 0.5°.

The four telescopes share a primary of $f/3$ ($R_1 = -1200\,\mathrm{mm}$) and a secondary magnification $m = 5$, giving

$f/15$ and an effective focal length of 3000 mm. The Newtonian instead uses a single $f/15$ paraboloid, $R = -6000\,\mathrm{mm}$. The conic constants and secondary radii follow from the design family, as listed in Table III.

IV. Results

A. Spherical and parabolic mirrors

Table II collects the spot data. On axis the spherical mirror reproduces (1) closely: the measured geometrical radius is 511.5 µm at $f/2.5$ against the 500 µm predicted, and 7.82 µm at $f/20$ against 7.81. The agreement improves from 2.3 % to 0.06 % as N grows, which is the higher-order content of the aberration disappearing.

The spherical mirror stops being the dominant defect near $f/15$: there the on-axis RMS radius, 7.96 µm, falls below the Airy radius of 10.07 µm, and at $f/20$ the whole geometrical spot fits inside the Airy disc. Figures 1 and 3 show the same mirror at the two extremes.

The paraboloid gives an on-axis RMS radius of zero at every focal ratio (Fig. 2). Off axis it is no better than the sphere: at 0.5° its RMS radius falls as $1/N$, from 56.1 µm at $f/2.5$ to 7.40 µm at $f/20$, which is the scaling of (2) at fixed field angle.

TABLE II: Single mirrors of 200 mm aperture. Radii in µm, from the Zemax spot diagrams.

N	Airy rad.	RMS 0°	GEO 0°	RMS 0.5°
Spherical, $K = 0$				
2.5	1.686	201.900	511.486	256.190
5	3.350	71.913	125.707	68.110
10	6.712	17.912	31.294	21.985
15	10.066	7.956	13.895	13.054
20	13.421	4.474	7.815	9.630
Parabolic, $K = -1$				
2.5	1.604	0.000	0.000	56.119
5	3.363	0.000	0.000	28.119
10	6.714	0.000	0.000	14.470
15	10.068	0.000	0.000	10.087
20	13.422	0.000	0.000	7.398

B. Two-mirror telescopes

All four designs null spherical aberration on axis: the RMS radius is below 0.01 µm in every case (Table IV), which confirms the conic constants of Table III. The differences appear entirely in the field.

At 0.5° the Newtonian gives 11.6 µm, the Gregorian 48.6, the Ritchey-Chrétien 64.6 and the classical Cassegrain 63.4, against an Airy radius of 10.07 µm. The Cassegrain and the Ritchey-Chrétien are indistinguishable in RMS, but their maximum radii differ, 115.1 against 96.8 µm. Fig. 4 shows the Cassegrain spot diagrams.

C. Lenses

Two N-BK7 singlets were traced, an equi-biconvex at $f/5$ and a plano-convex at $f/2.5$ (Table V, Fig. 5). Their on-axis RMS radii, 478 and 860 µm, are two orders of magnitude above the Airy radius, and the spot barely changes between the on-axis and the 1° field, which shows the residual is spherical aberration and not a field aberration. Compared with the spherical mirror of the same focal ratio the singlet is between four and seven times worse, and it also carries the chromatic spread of the three wavelengths, absent in every mirror system.

TABLE III: Prescriptions of the four telescopes, all 200 mm and $f/15$.

Design	R_1 [mm]	K_1	R_2 [mm]	K_2	Obstr. [%]
Newtonian	−6000	−1	—	—	7.5
Classical Cassegrain	−1200	−1	−333.33	−2.2500	22.2
Gregorian	−1200	−1	+333.33	−0.4444	33.3
Ritchey-Chrétien	−1200	−1.0234	−333.33	−2.4561	22.2

TABLE IV: RMS spot radius of the telescopes against field angle, in µm. The Airy radius at $f/15$ is 10.07 µm.

Design	0°	0.25°	0.5°	Max 0.5°
Newtonian	0.000	5.54	**11.64**	29.45
Gregorian	0.002	13.06	48.57	77.37
Classical Cassegrain	0.002	16.55	63.37	115.1
Ritchey-Chrétien	0.002	16.12	64.55	**96.78**

TABLE V: N-BK7 singlets of 200 mm aperture, traced at 450, 550 and 650 nm. Radii in µm.

Shape	N	RMS 0°	Max 0°	RMS 1°	Mirror, same N
Equi-biconvex	4.9	478.3	1106	426.0	71.9
Plano-convex	2.44	860.3	1903	702.4	201.9

V. Discussion

A. Mirrors

The paraboloid removes spherical aberration exactly and at every focal ratio, but leaves the field untouched, exactly as (2) predicts since coma does not depend on K. The sphere is the opposite case: its defect is entirely on axis and disappears on its own as the focal ratio grows, because W_{PV} falls as N^{-3}. Past $f/15$ the two surfaces are equivalent for practical purposes, which is the reason spherical mirrors are still used.

B. Telescopes

Nulling spherical aberration is achieved by all four designs and is therefore not a discriminant. What separates them is the field, and the ordering is not the expected one: the Newtonian is five times better at 0.5° than the two Cassegrain-type designs. The reason is that the Newtonian is a single slow $f/15$ surface, while the others fold a fast $f/3$ primary whose field aberrations the secondary amplifies. The Newtonian pays for it with a three-metre tube against 467 mm, and the Gregorian pays for its flatter field with an 800 mm tube and a 33 % obstruction.

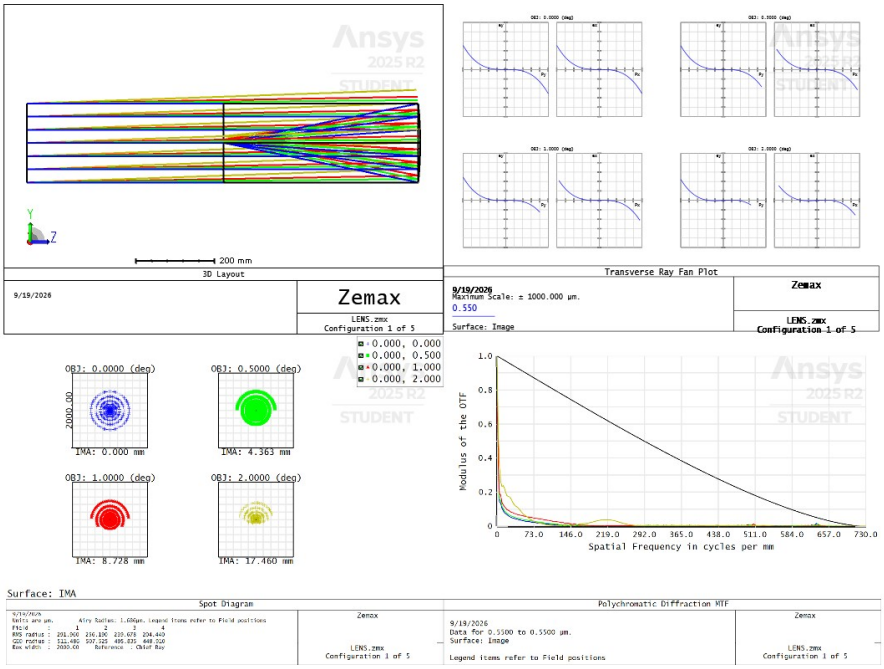


Fig. 1: Spherical mirror at $f/2.5$: layout, transverse ray fan, spot diagram and polychromatic diffraction MTF. The ray fan shows the cubic signature of spherical aberration and the MTF collapses within a few tens of cycles per millimetre.

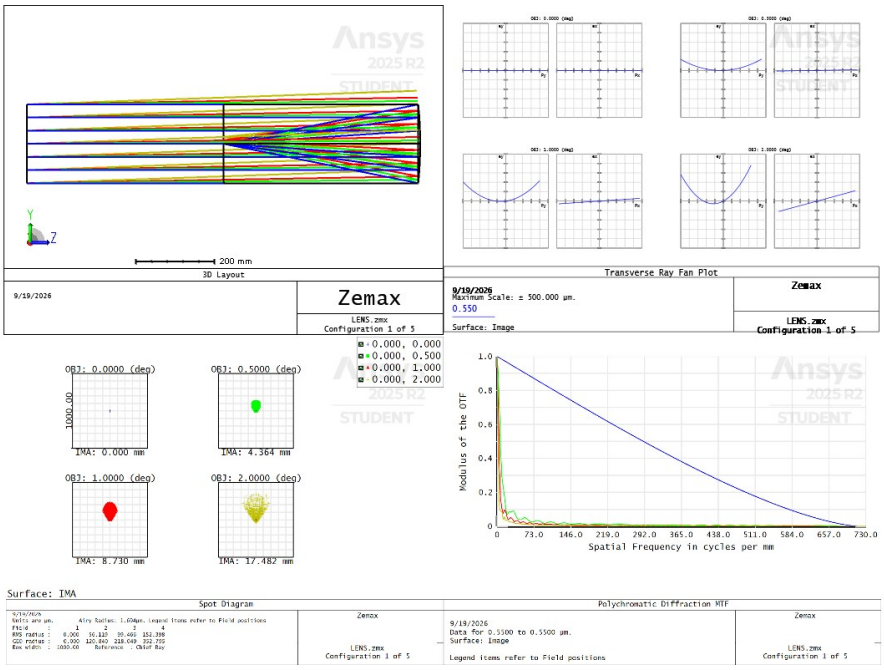


Fig. 2: Parabolic mirror at $f/2.5$. The on-axis spot collapses to a point and the on-axis MTF follows the ideal curve; the off-axis fields keep the coma that the conic constant cannot remove.

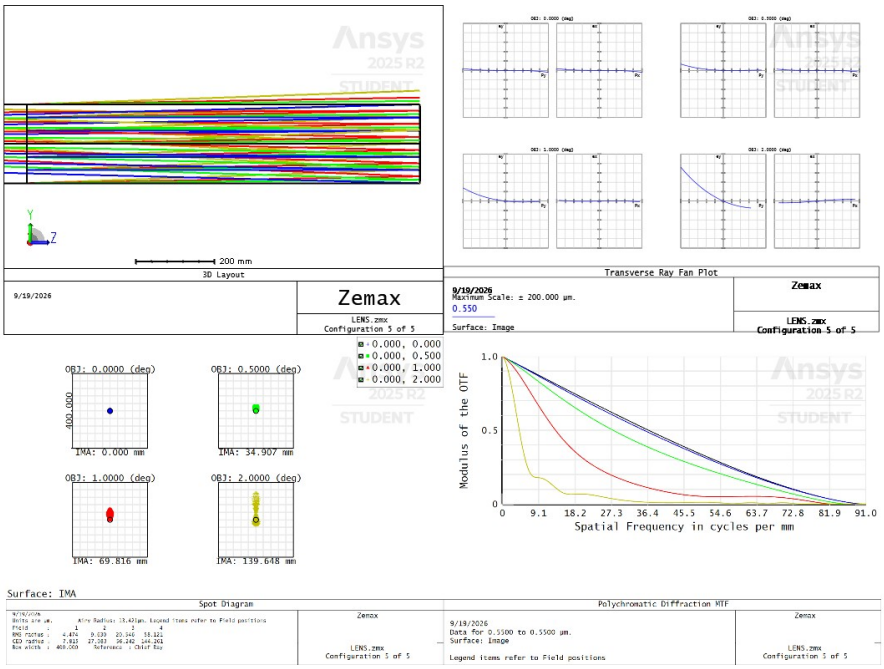


Fig. 3: Spherical mirror at $f/20$. The geometrical spot now fits inside the Airy disc and the MTF approaches the diffraction limit, so at this focal ratio the sphere behaves as a perfect surface.

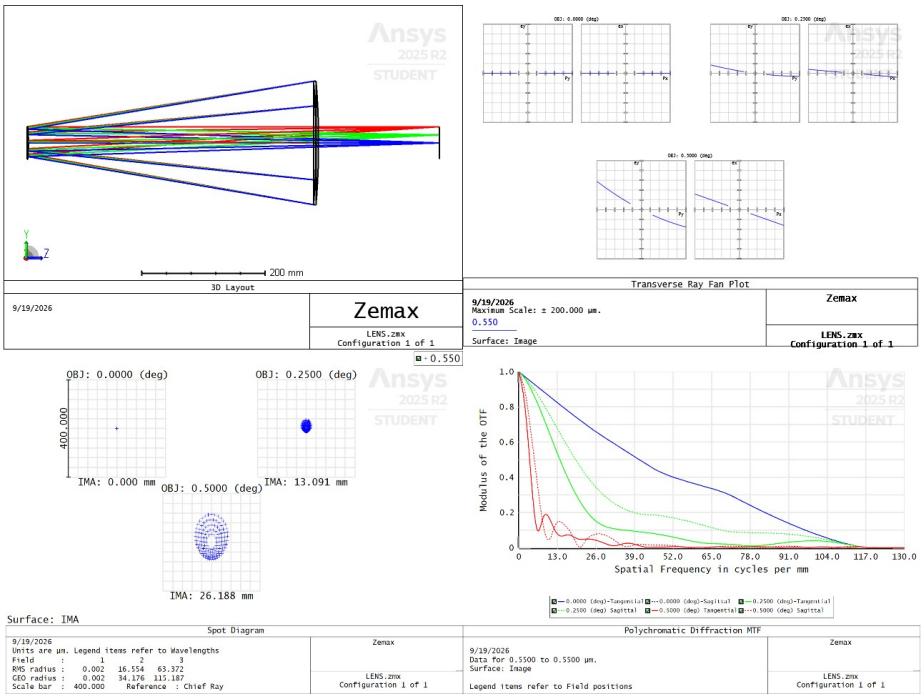


Fig. 4: Classical Cassegrain, $f/15$. The on-axis spot is a point; the off-axis spots grow quadratically with the field angle.

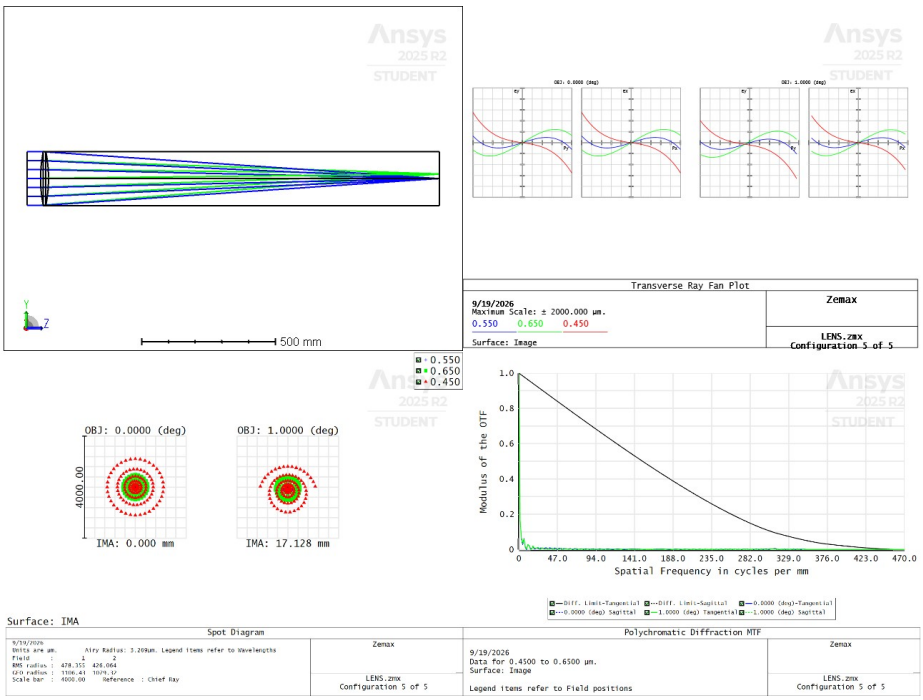


Fig. 5: Equi-biconvex N-BK7 singlet. The three wavelengths separate both in the ray fan and in the spot diagram, and the MTF is close to zero over almost the whole frequency range.

The Ritchey-Chrétien does not improve the RMS radius over the classical Cassegrain, which at first sight contradicts its purpose. It does not: at 0.5° the dominant aberration is already astigmatism, which the extra conic constant does not touch. The evidence is the scaling, since the RMS grows by a factor of four between 0.25 and 0.5°, which is the β^2 law of astigmatism and not the β law of coma. The coma that the Ritchey-Chrétien does remove appears in the maximum radius, 16 % smaller, because it is the asymmetric tail of the patch and not its overall size.

C. Lenses

The singlets are the worst systems measured, between four and seven times worse than a mirror of the same focal ratio and with a chromatic spread on top of it. Both defects come from the same place: refraction depends on the index, which is both strongly curvature-dependent and wavelength-dependent, whereas reflection is neither. This is the quantitative reason why every large astronomical telescope is a reflector.

VI. Conclusions

Spherical aberration is a solved problem in two independent ways: by aspherizing the surface, which works at any focal ratio, or by making the system slow enough, since the wavefront error falls as N^{-3} and the $f/20$ sphere is already diffraction limited. Neither solution helps off axis.

Among the telescopes the field, not the axis, is what separates the designs, and the aberration that limits them at 0.5° is astigmatism rather than coma. That is why the Ritchey-Chrétien shows its advantage only in the shape of the patch, and why the slow single-mirror Newtonian, at the cost of a tube six times longer, outperforms all the folded designs.

The main limitation of this analysis is that the two singlets were traced at different focal ratios, so the effect of the lens shape could not be isolated from the effect of N; repeating both shapes at a common focal ratio, and adding the achromatic doublet, would complete the refracting section.

References

[1] L. Vanzi, *Optical Aberrations* and *Computer Analysis and Design of Optical Systems for Astronomy*, course notes IEE3873, Pontificia Universidad Católica de Chile.

[2] M. Riccardi, *ATMOS – Optical Design and Analysis Software*, v8.96. https://www.atmos-software.it

[3] D. J. Schroeder, *Astronomical Optics*, 2nd ed. San Diego, CA: Academic Press, 2000.